

Economics and Accounts for Engineers. ①

- Economics is originated from a greek word 'OIKONOMIKOS' which means house management.
- Economics explains how different individuals behave while managing their economic activities.
- Every Economic system irrespective of its nature has to face three kinds of economic problems.

1. What to produce in what quantities?

Every Economic system has to decide whether the resources have to be allocated for the production of consumer goods or for the production of producer goods (capital goods).

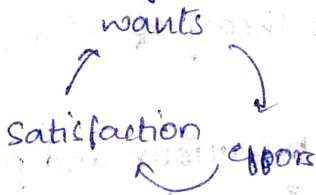
2. How to produce?

What methods are to be used for the production i.e., whether we use capital intensive techniques or labour intensive techniques.

3. For whom to produce?

Problems related to distribution the total goods & services produced in economy are to be distributed among the factors which contribute to the production.

→ Relationship among wants, efforts and satisfaction.



- When want arises it is satisfied by making an effort, as soon as one want is satisfied another want arises effort is made again to satisfy this want.
- So the process of wants, efforts and satisfaction continues.

Definitions of Economics

① Wealth definition:

It is given by Scottish economist 'Adam Smith' is also father of economic. According to him economics is the study of nature and uses of national wealth.

② Welfare definition:-

It is given by 'Dr. Alfred Marshall', according to him economics is the study of man's actions in the ordinary business of life. It acquires how he gets his income and how he uses it.

③ Scarcity definition:-

It is given by 'Professor Lionel Robbins', according to him economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses.

④ Growth definition:-

It is given by 'Paul A. Samuelson', according to him economics is the study of how people and society end up choosing with and without use of money to employ scarce productive resources that could have alternative uses to produce various commodities and distribute them for consumption now or in the future among various persons and groups in the society.

Micro economics:-

Study of a individual customer or a firm is called micro economics. It is also called as theory of firm.

Micro means 1 million

Micro economics deals with behaviour and problem of individual and of microorganisation.

Macro Economics:-

The study of Aggregate or total level of Economic activity in a activity is called Macro economics.

It studies the flow of Economic resources from resource owner to the business firms and business firms to the households.

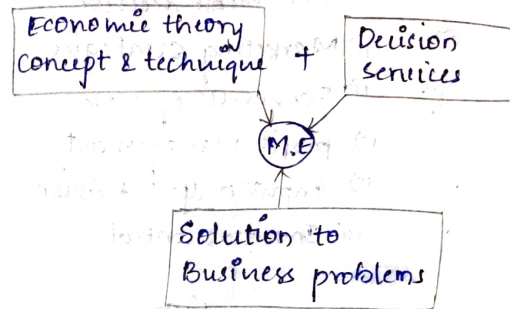
It deals with total aggregate for instance total national income, total consumption, total Employment etc.

②

→ According to spence & siegelman, ME is an Integration of Economic theory with an business practice for the purpose of facilitating decision making and formed planning by management

→ According to Brigham & papper, ME is an application of economic theory and methodology to Business Administration practice.

→ According to Haqus &, ME is the fundamental Academic subject which seeks to understand and to analyse problem of Business decision making.



Nature of Managerial Economics

→ ME is the youngest of all social sciences since it originate from economics, it has a basic features of economics such as, assuming that things remain the same (Latin equivalent word ceteris paribus)

- 1) Close to micro economics
- 2) Operates against the back drop of Macro economics
- 3) Normative statements
- 4) Prescriptive actions
- 5) Applied Nature
- 6) Offers scope to evaluate each alternative
- 7) Inter disciplinary
- 8) Assumptions & limitations.

Scope of ME:-

The main focus of ME is to find optimal solutions to a given managerial problems.

Concepts & Techniques of ME

Applied to

- ② Demand analysis & Demand forecasting
- ③ Resource planning
- ④ competitive analysis
- ⑤ strategic planning
- ⑥ Production management
- ⑦ cost analysis
- ⑧ Pricing strategies
- ⑨ Market structure analysis
- ⑩ Investment decision
- ⑪ Marketing strategies
- ⑫ Economics of scale
- ⑬ profit management
- ⑭ Input output decisions
- ⑮ Investors control.

given

Optimal solutions.

Linkages with other disciplines

- 1) Economics
- 2) Operations
- 3) Mathematics
- 4) Statistics
- 5) Accountancy
- 6) Psychology
- 7) Organizational behaviour.

willingness

What is demand?

Every want supported by the ~~best interest~~ and ability to buy constitutes demand for a particular product or service.

A product or service is said to have a demand if it satisfies three conditions:

- ① Desire on the part of the buyer to buy

5

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② willingness to pay

③ Ability to pay specified price for it.

Nature and types of demand.

1. Consumer goods vs producer goods
2. Autonomous demand vs derived demand.
3. Durable vs perishable goods
4. Firm demand vs industry demand.
5. Short run vs long run demand
6. New demand vs replacement demand.
7. Total market demand vs segment market demand.

Demand Determinance.

The demand for a particular product depends on several factors. They are.

- ① Price of commodity - inversely proportional (P)
- ② Income levels of the consumers - directly (I)
- ③ Tastes and preferences of the consumer. (T)
- ④ Prices of related goods which may be substitutes or complements (P_e)

Price (P)
Income (I)

- ⑤ Expectations about the prices in the future (E_P)
- ⑥ Expectations about the incomes in the future (E_I)
- ⑦ size of population (S_P)
- ⑧ Distribution of consumers over different regions. (D_e)
- ⑨ Advertising efforts (A_e)
- ⑩ Any other factor capable of affecting the demand. (P_e)

Demand Function

$$Q_d = f(P, I, T, P_e, E_P, E_I, S_P, D_e, A_e, P_e)$$

(demand determinance)

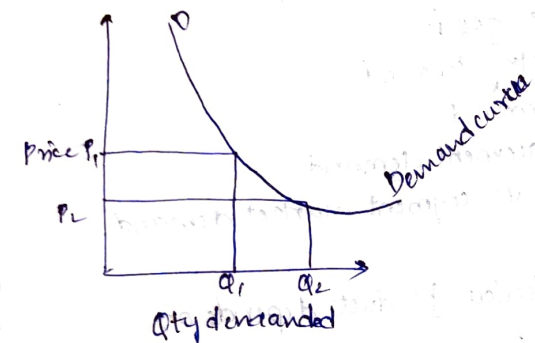
$$-\frac{P}{Q_d} \cdot \frac{Q_d}{P} = -\frac{P}{Q_d} \cdot \frac{Q_d}{P} = -1$$

⑥

Law of Demand

The amount of quantity demanded rises with every fall in the price and vice versa when other things remain the same.

Other things remain the same includes $I, T, P_r, E_p, E_c, S, D, M, O$



Exceptions:-

- ① Where there is a shortage of necessity ~~spheres~~ ^{feared}
- ② Where the product is such that it confers distinction. (Veblen goods)

Giffen's Paradox

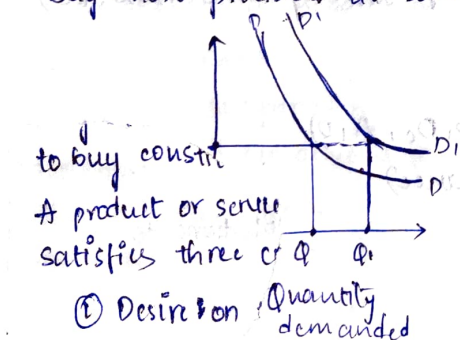
- ④ In case of ignorance of price track.

Change in Demand

Increased or decrease in demand due to the change in the factors other than price is called change in demand.

Increase in Demand

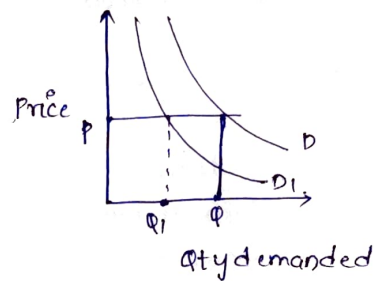
It occurs when buyers are willing and able to buy more products at same price.



- ① Desire to buy

Decrease in demand

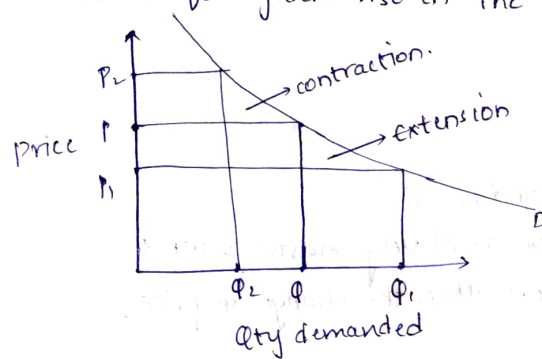
It occurs when buyers are ready to buy less of a product at same price.



Extension and contraction in demand

An extension is a downward movement along a demand curve which indicates that a higher quantity is demanded for a given fall in the price of a commodity.

A contraction is an upward movement along a demand curve which indicates that the lower quantity is demanded for given rise in the price of commodity.



Elasticity of demand

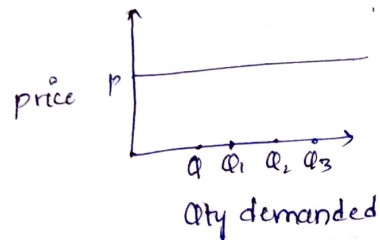
Def:- The rate of responsiveness to the quantity demand of a commodity for a given change in the price or any other demand determinants.

Measurements of elasticity of demand $e = \frac{Q_2 - Q_1}{Q_1} \times \frac{P_1}{P_2 - P_1}$

- ① Perfectly elastic demand
- ② Perfectly inelastic demand
- ③ Relatively elastic demand
- ④ Relatively inelastic demand
- ⑤ Unity elasticity of demand.

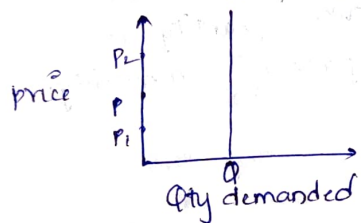
$$e = \frac{\Delta Q}{Q_1} \times \frac{P_1}{\Delta P} = ⑥$$

- ① Perfectly elastic demand
When any quantity can be sold at a given price there is no need to reduce price then it is said to be perfectly elastic demand.
($e = \infty$)



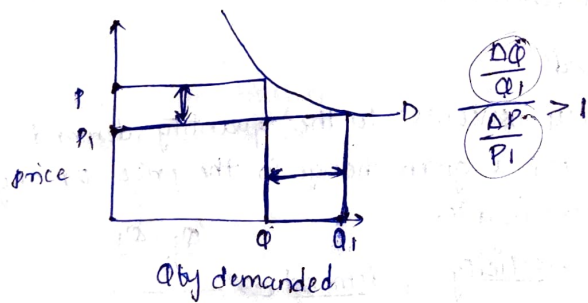
② Perfectly inelastic demand

The demand is said to be perfectly inelastic when there is no change in the quantity demanded even though there is a big change in price.
($e = 0$)



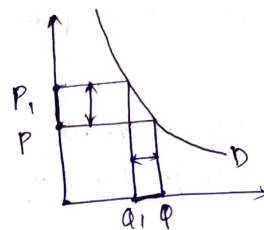
③ Relatively elastic demand ($e > 1$)

The demand is said to be relatively elastic when the change in demand is more than the change in price.



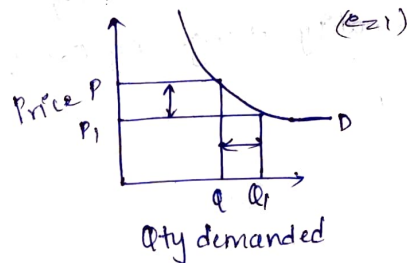
④ Relatively inelastic demand ($e < 1$)

The demand is said to be relatively inelastic when the change in demand is less than change in price.



⑤ Unity elasticity of demand

The demand is said to be unity if change in demand is equal to change in price.



Here '-' sign represents inverse Relationship between Price and quantity demanded.

Interpretation:- For every 1% change in price of commodity, the quantity demanded of a commodity changes by 5%.

Case 2:- Inelastic price demand ($E < 1$)

$$E_{dp} = -0.33$$

Interpretation:- For every 1% change in price of commodity, the quantity demanded by a commodity changes by 0.33%.

Case 3:- Unity price elasticity ($E = 1$)

$$Q_1 = 1000 \quad P_1 = ₹ 100$$

$$Q_2 = 1500 \quad P_2 = ₹ 50$$

$$E_{dp} = \frac{\frac{500}{1000}}{\frac{-50}{100}} = \frac{\frac{1}{2}}{-\frac{1}{2}} = -1$$

Price elasticity

Interpretation:- For every 1% change in price of a commodity, the quantity demanded by a commodity changes by 1%.

Type 2:- Income elasticity of demand

It refers to the quantity demanded of a commodity in response to given change in the income levels of the consumer.

Income elasticity of demand = $\frac{\text{Proportionate change in the quantity demanded}}{\text{Proportionate change in the income of consumer}}$

$$E_{di} = \frac{Q_2 - Q_1}{Q_1} \div \frac{I_2 - I_1}{I_1}$$

I_1, I_2 are income of consumer before and after change.

Case 1:- Elastic income demand ($E > 1$)

$$Q_1 = 1000 \text{ units} \quad I_1 = ₹ 100$$

$$Q_2 = 1400 \text{ units} \quad I_2 = ₹ 80$$

$$E_{di} = \frac{\frac{-300}{1000}}{\frac{-20}{100}} = \frac{3}{10} \times 5 = 1.5$$

Interpretation:- For every 1% change in income of consumer, the quantity demanded changes by 1.5%.

Case 2:- Inelastic income demand ($E < 1$)

$$Q_1 = 1500 \text{ units} \quad I_1 = ₹ 100$$

$$Q_2 = 1000 \text{ units} \quad I_2 = ₹ 50$$

$$E_{di} = \frac{\frac{-500}{1500}}{\frac{-50}{100}} = \frac{-\frac{1}{3}}{-\frac{1}{2}} = \frac{2}{3} = 0.66$$

For every 1% change in income of consumer, the quantity demanded of the commodity changes by 0.67%.

Case 3:- Unity Income elasticity ($E = 1$)

$$Q_1 = 2000 \quad I_1 = 100 ₹$$

$$Q_2 = 1000 \quad I_2 = 50 ₹$$

$$E_{di} = \frac{\frac{-1000}{2000}}{\frac{-50}{100}} = \frac{-\frac{1}{2}}{-\frac{1}{2}} = 1$$

For every 1% change in income of consumer, the quantity demanded of a commodity changes exactly by 1%.

Type 3:- Cross elasticity of demand.

It refers to the quantity demanded of a commodity in response to change in the prices of related goods which may be substitutes or complements.

Cross elasticity of demand = $\frac{\text{Proportionate change in quantity demanded for a commodity}}{\text{Proportionate change in price of commodity Y}}$

$$E_{dc} = \frac{Q_2 x - Q_1 x}{Q_1 x} \cdot \frac{P_2 y - P_1 y}{P_1 y}$$

Case 1:- Elastic cross demand ($e > 1$)

$$Q_1 x = 1000 \quad P_1 y = \text{Rs } 20 \\ Q_2 x = 2000 \quad P_2 y = \text{Rs } 30$$

$$E_{dc} = \frac{1000}{1000} \cdot \frac{10}{20} = 2$$

for every 1% change in price of commodity 'y' the quantity demanded for commodity 'x' changes by 2%.

Case 2:- Inelastic cross demand ($e < 1$)

$$Q_1 x = 1000 \quad P_1 y = 20 \\ Q_2 x = 1200 \quad P_2 y = 30$$

$$E_{dc} = \frac{200}{1000} \cdot \frac{10}{20} = \frac{2}{5} = 0.4 < 1$$

1% change in 'y', qty demand in 'x' change by 0.4%.

Case 3:- Unity cross demand ($e = 1$)

$$Q_1 x = 1000 \quad P_1 y = 20 \\ Q_2 x = 1500 \quad P_2 y = 30$$

$$E_{dc} = \frac{500}{1000} \cdot \frac{10}{20} = 1$$

1% change in 'y', qty demand in 'x' change by 1%.

Type 4:- Advertising elasticity demand
It refers to increase in sales bcoz of change in advertising expenditure.

$$AED = E_{da} = \frac{Q_2 - Q_1}{Q_1} \cdot \frac{A_2 - A_1}{A_1}$$

Advertising exp before
& after change

Case 1:- Elastic advertising demand ($e > 1$)

$$Q_1 = 100000 \quad P \cdot A_1 = 10000 \\ Q_2 = 30000 \quad A_2 = 5000$$

$$E_{da} = \frac{-100000}{100000} \cdot \frac{-5000}{10000} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 0.25$$

1% change in 'A' = 0.25%

Case 2:- Inelastic advertising demand ($e < 1$)

$$Q_1 = 100000 \quad A_1 = 5000 \\ Q_2 = 30000 \quad A_2 = 1500$$

$$E_{da} = \frac{-20000}{100000} \cdot \frac{-3500}{5000} = \frac{1}{5} \cdot \frac{7}{10} = \frac{7}{50} = 0.14$$

1% change in 'A' = 0.14%

Case 3:- $e = 1$

$$Q_1 = 50 \quad A_1 = 5000 \\ Q_2 = 25 \quad A_2 = 2500$$

$$E_{da} = \frac{-25}{50} \cdot \frac{-2500}{5000} = 1$$

1% change in 'A' = 1%.

Factors governing elasticity of demand.

Nature of products.

Time frame

Degree of postponement

Number of alternative uses

Tastes & preferences of a consumer

availability of ~~close~~ substitutes

level of prices.

Availability of substitutes

Durability of product

Government policy

Significance

the concept of elasticity is very useful to the producers and policy makers alike.

It is very valuable tool to decide the extent of $\uparrow$ or $\downarrow$ in price for a desired change in the quantity demanded of the product or service in the or an economy.

Application:-

- ① To fix the prices of factors of production.
- ② To fix the prices of goods & services provided or rendered.
- ③ Formulate or revise government policies.
- ④ Forecasting demand
- ⑤ Planning the levels of output & price.

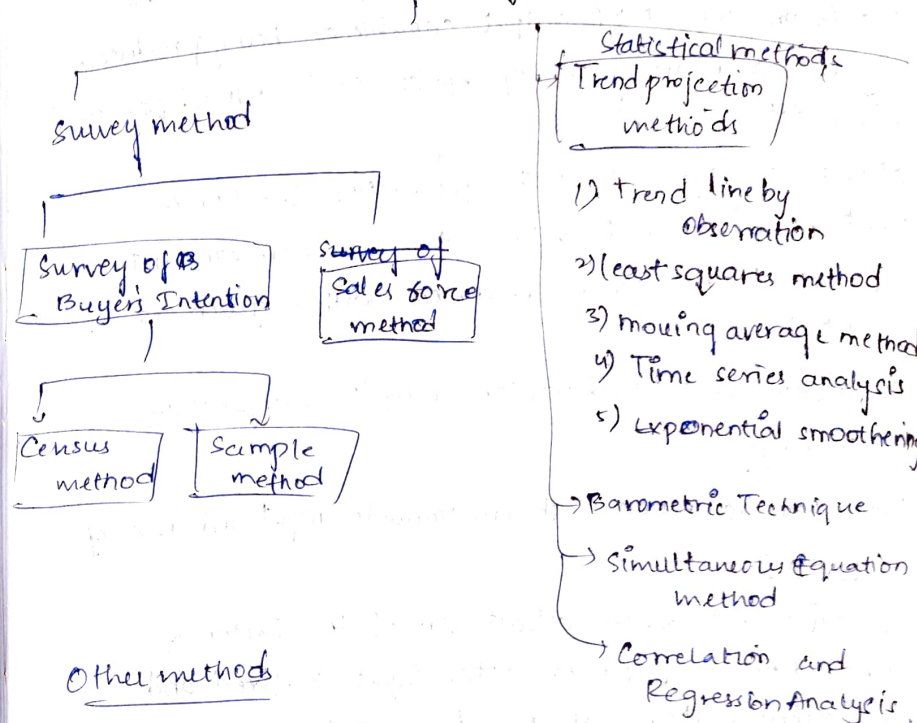
Demand forecasting

measure

It is necessary to demand accurately in terms of quantity and its value for several purposes. Forecasting helps to assess the likely demand for the products & services and to plan production accordingly.

Demand forecasting is helpful not only at firm level but also at national level.

Methods of Demand Forecasting



Other methods

- ① Expert opinion method
- ② Test marketing
- ③ Controlled experiment
- ④ Judgemental Approach.

Factors governing demand forecasting.

- ① Internal factors
 - ① Pricing strategies
 - ② marketing & sales initiative
 - ③ Product lifecycle
 - ④ company capacity
 - ⑤ New product launches
- ② External factors
 - ① economic condition
 - ② market trends
 - ③ competitive analysis
 - ④ seasonality
 - ⑤ Regulatory changes
 - ⑥ Technological factors
 - ⑦ Innovation
 - ⑧ Automation in supply chain
 - ⑨ Political instability
 - ⑩ Natural disasters
 - ⑪ Historical events
 - ⑫ Past sales
 - ⑬ External forces or prices
 - ⑭ Past sales, external historical events
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Theory of production

Production Function:- Samuelson defined the production function as the technical relationship which reveals maximum amount of output capable of being produced by each and every set of inputs. It is defined for a given state of technical knowledge.

Michael R Baye defines the production function as that function which defines the maximum amount of output that can be produced with a given set of inputs.

$Q = f(L, C, K, O, T)$ [Land, labour, capital, organisation & Technology]

Production function with one variable input and law of returns.

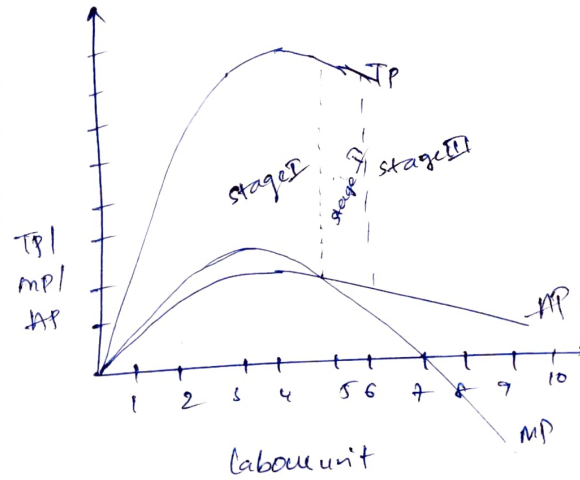
It states that when atleast one of the factor of production is fixed or factor input is fixed and all other factors are varied, the total output per unit of labour initially, after reaching a certain labour unit of output the total output will increase at declining rate. if variable factors are input added further to the fixed factor unit the total output may decline.

The law of return is also called law of variable proportion (LVP)

The laws of diminishing returns.

Labour unit	Total product (TP)	Marginal product	Average product
0	0	0	0
1	10	10	10
2	22	12	11
3	33	11	11

4	40	7	stage II	10
5	45	5		9
6	48	3	stage III	8
7	46	0		6.6
8	45	-3		5.6



Returns to scale :-

It refers to returns or benefit enjoyed by the firm as a result of change in all the inputs.

Law of increasing returns to scale :-

This law states that the volume of output keeps on increasing with every increase in inputs where a given increase in inputs leads to more than the proportionate increase in output.

Law of constant returns to scale :-

This law states that the rate of increase or decrease in volume of output is same to that of the rate of increase or decrease in input.

Law of decreasing returns to scale :-

This law states that when the proportionate increase in input does not lead to equivalent increase in output. The output increases at a decreasing rate.

Capital Labour	% increase in input	G.P.P	% increase in output	Law of application
1	3	50	-	-
2	6	100	100%	Increasing return to scale
4	12	200	100%	constant return to scale
8	24	360	50%	decreasing return to scale

Returns to factors :-

Returns to factors refers to output or return generated as a result of change in 1 or more factors keeping the other factors unchanged.

Economies and Diseconomies of scale :-

The economies of scale result because of increase in the scale of production. Dr. Alfred Marshall divide the economies of scale into 2 groups :-

- Internal economies
- External economies

(i) Internal economies

It refers to the economies in the production cost which accrue to the firm alone.

- Managerial economies
- Commercial economies
- Financial economies
- Technical economies
- Marketing economies
- Risk bearing economies
- Indivisible and automated machinery
- Economies of larger dimensions
- Economies of research & development.

(ii) External economies

It refers to the benefits which accrue to all the firms in the industry.

- Economies of concentration
- Economies of Research & Development
- Economies of welfare

Diseconomies of scale

Diseconomies are mostly managerial in nature. Problems of planning, coordination, communication & control may become increasingly complex as the firm grows in size resulting in increasing in avg cost per unit. Sometimes the firm may also collapse.

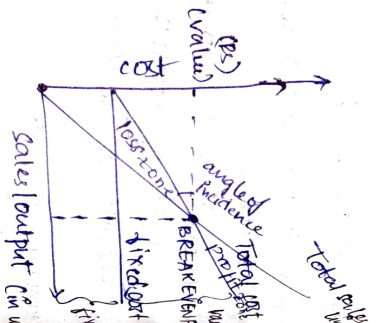
Cost Analysis

Cost is defined as the amount of expenditure income actual or notional to produce a particular product or service.

Various cost concepts :-

- Long run versus short run cost
- Fixed cost vs variable cost
- Marginal cost vs
- Semi variable ~~or~~ semi fixed
- Controllable vs uncontrollable cost
- Opportunity cost vs outlay cost
- Incremental cost vs sunk cost
- Implicit cost vs explicit cost
- Out of pocket cost vs book cost
- Replacement cost vs historical cost
- Past cost vs future cost
- Separable cost vs joint cost
- Urgent vs postponable cost
- Escapable vs unavoidable cost

② Long run vs short run Break even Analysis :-



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Break Even Analysis.

$$\text{Selling price} = \text{cost} + \text{profit} \Rightarrow S = C + P$$

$$\text{Selling price} = \text{fixed cost} + \text{variable cost} + \text{profit} \Rightarrow S = F + V + P$$

$$\text{Sales} - \text{variable cost} = \text{fixed cost} + \text{profit} = \text{Contribution}$$

$$\text{Contribution} = S - V = F + P$$

$$\text{Contribution per unit} = S_{pu} - V_{pu}$$

$$= \text{sales per unit} - \text{variable cost per unit}$$

$$\frac{\text{Profit}}{\text{volume}} \text{ ratio} = \frac{P}{V} \text{ Ratio} = \frac{\text{Contribution}}{\text{Sales}}$$

$$\text{Break even point (in volume)} = \frac{\text{Fixed cost}}{\frac{P}{V} \text{ Ratio}} = \frac{F}{\left(\frac{P}{V}\right)}$$

$$\text{Break even point (in units)} = \frac{\text{Fixed cost}}{\text{contribution per unit}}$$

$$\text{margin of safety (MOS)} = \text{Total sales} - \text{B.E.P}$$

$$\text{Sales required to earn a desired target profit} = \frac{\text{Fixed cost} + \text{desired profit}}{\frac{P}{V} \text{ ratio}}$$

$$\text{Fixed cost} + \text{desired profit} / \left(\frac{P}{V}\right)$$

$$\text{No. of units to be manufactured and sold} =$$

$$\frac{\text{Fixed cost} + \text{desired profit}}{\text{contribution per unit}}$$

$$\frac{P}{V} = \frac{\text{change in profits}}{\text{change in sales}}$$

Problems

① Find the amount of variable cost from the following information.

Sales Rs 285,000, Fixed cost = 50,000, Profit = Rs 60,000

Ans:- we know that,

$$\text{Sales} - \text{variable cost} = \text{Fixed cost} + \text{profit}$$

$$285,000 - V = 50,000 + 60,000$$

$$V = 285,000 - 110,000 = \underline{175,000} \text{ Rs } 1,75,000.$$

② Calculate BEP from the following figures.

sales = Rs 3,00,000

Fixed expenses = Rs 75,000

direct material = Rs 100,000

direct labour = Rs 60,000

direct expenses = Rs 40,000

$$\text{Ans:- B.E.P (in volume)} = \frac{\text{Fixed cost}}{\left(\frac{P}{V}\right) \text{ ratio}}$$

$$\frac{P}{V} \text{ ratio} = \frac{\text{contribution}}{\text{sales}} = \frac{S - V(\text{DM} + \text{DL} + \text{DE})}{\text{sales}}$$

$$\frac{3,00,000 - (2,00,000)}{300,000} = \frac{1,00,000}{3,00,000} = \frac{1}{3}$$

$$\text{BEP (in volume)} = \frac{75,000}{\left(\frac{1}{3}\right)} = \text{Rs } 2,25,000$$

75
1/3
225

③ A company's turnover in a year was Rs 50,00,000 and its profit Rs 5,00,000. Its P/V ratio was 40%. What was the BEP.

Ans:- Sales = Rs 50,00,000

Profit = Rs 5,00,000

$$\frac{P}{V} = 40\% = \frac{40}{100} = \frac{2}{5}$$

$$\text{BEP (in volume)} = \frac{F}{\left(\frac{P}{V}\right)}$$

$$\frac{P}{V} = \frac{\text{contribution}}{\text{sales}}$$

$$\frac{2}{5} = \frac{\text{contribution}}{50,00,000}$$

$$\text{contribution} = 20,00,000$$

$$\text{contribution} = F + P$$

$$20,00,000 = F + 5,00,000$$

$$F = 15,00,000$$

$$\text{B.E.P (in volume)} = \frac{F}{\left(\frac{P}{V}\right)} = \frac{15,00,000}{\left(\frac{2}{5}\right)} = \frac{37,50,000}{2} = 37,50,000 \text{ Rs}$$

Q From the following figures calculate sales required to earn a profit of Rs. 1,20,000.

Sales Rs 6,00,000

V.C Rs 3,75,000

F.C Rs 1,80,000

Sales Required to earn a desired profit =

$$\frac{\text{Fixed cost} + \text{desired profit}}{\left(\frac{P}{V}\right)}$$

$$\frac{P}{V} = \frac{\text{Contribution}}{\text{Sales}} = \frac{S - V}{S} = \frac{6,00,000 - 3,75,000}{6,00,000}$$

$$= \frac{2,25,000}{6,00,000} = \frac{45}{120} = \frac{3}{8}$$

$$\Rightarrow \frac{1,80,000 + 1,20,000}{\left(\frac{3}{8}\right)} = \frac{3,00,000}{\frac{3}{8}} = 8,00,000 \text{ Rs}$$

Q What would be the volume of sales to derive a profit of Rs 20,000. If the P/V ratio is $66\frac{2}{3}\%$ and fixed ~~overhead~~ ^{indirect expenses} for the period Rs 40,000.

Ans:

$$\frac{P}{V} \text{ Ratio} = 66\frac{2}{3}\% = \frac{200}{3}\% = \frac{200}{3 \times 100} = \frac{2}{3}$$

Sales required to earn a desired profit of Rs 20,000

$$\begin{aligned} \frac{F + P}{\left(\frac{P}{V}\right)} &= \frac{40,000 + 20,000}{\frac{2}{3}} = \frac{60,000 \times 3}{2} \\ &= 90,000 \end{aligned}$$

Q The following data is given,

fixed expenses = Rs 10,00,000

V.C = Rs 10/unit

S.P = sales = Rs 15/unit

Indicate the no. of units to be manufactured and sold.

i) To Break even

ii) To earn a profit of Rs 10,000

iii) What additional units would be necessary to increase the above profit by Rs 5,000

$$\text{Ans: B.E.P (in units)} = \frac{\text{Fixed cost}}{\text{contribution per unit}}$$

$$\begin{aligned} &= \frac{10,00,000}{\text{Selling price} - \text{V.C}} = \frac{10,00,000}{15 - 10} \\ &= \frac{2,00,00,000}{5} = 2,00,000 \text{ units.} \end{aligned}$$

No. of units to be manufactured
To earn a profit Rs 10,000 = $\frac{\text{Fixed cost} + \text{desired cost}}{\text{contribution per unit}}$

$$= \frac{10,00,000 + 10,000}{5}$$

$$= \frac{10,00,000}{5} = 2,00,000 \text{ units}$$

No. of units to be manufactured to increase the above profit by Rs 5000 i.e., $10,00,000 + 5,000 = 15,000$ (new)

$$\frac{\text{Fixed cost} + \text{D.C.}}{\text{CPU}} = \frac{2,20,000}{5} = 44,000 \text{ Rs}$$

Additional units required =

2,03,000 units

④ From the following data calculate B.E.P.

Selling price per unit = Rs 20

Direct material cost per unit = Rs 8

Direct labour cost per unit = Rs 2

Direct expenses per unit = Rs 2

Variable overhead per unit = Rs 3

Fixed overhead Rs = 20,000

If sales are 20% above B.E.P. determine Net profit.

$$\text{B.E.P.} = \frac{\text{F.C.}}{\text{Contribution per unit}}$$

$$\text{contribution per unit} = \text{Selling price per unit} - \text{V.C. per unit (m, l, E, O)}$$

$$\text{B.E.P.} = \frac{20,000}{5} = 4,000 \text{ units}$$

New sales = 20% above B.E.P.

= 120% of B.E.P.

$$= \frac{120}{100} \times 4,000 = 4,800 \text{ units}$$

$$S - V = F + P$$

$$4,800 \times 20 - 4,800 \times 15 = 20,000 + \text{profit}$$

$$P = 96,000 - 72,000 - 20,000 = 4,000 \text{ Rs}$$

Profit + Fixed cost = contribution

$$P = (4,800 \times 5) - (20,000)$$

$$= 24,000 - 20,000 = 4,000 \text{ Rs}$$

⑤ The following data is available from the records of a company.

Sales Rs 1,00,000

V.C Rs 60,000

F.C Rs 20,000

You are required to calculate $\frac{P}{V}$, B.E.P. and margin of safety, also study the impact of change in the following variable on $\frac{P}{V}$ ratio, B.E.P. and margin of sales.

i) Increase in selling price by 10%.

ii) Decrease in fixed cost by 5000 Rs

$$\text{Ans: } \frac{P}{V} = \frac{\text{Contribution}}{\text{Sales}} = \frac{S - V}{S} = \frac{40,000}{1,00,000} = \frac{2}{5}$$

$$\text{B.E.P.} = \frac{\text{Fixed cost}}{\left(\frac{P}{V}\right)} = \frac{20,000}{\left(\frac{2}{5}\right)} = 50,000 \text{ Rs}$$

margin of safety = Total sales - B.E.P.

$$= 1,00,000 - 50,000$$

$$= 50,000 \text{ Rs}$$

① Increase in sales by 10%
 New sales = $1,00,000 \times \frac{110}{100}$
 $= 1,10,000 \text{ Rs} = \frac{110}{100} \times 1,00,000 = 1,10,000 \text{ Rs}$

i) $\frac{P}{V} \text{ ratio} = \frac{\text{contribution}}{\text{Sales}} = \frac{S-V}{S} = \frac{1,10,000 - 60,000}{1,10,000}$
 $= \frac{50,000}{1,10,000} = \frac{5}{11}$

ii) $BEP = \frac{\text{Fixed cost}}{(\frac{P}{V})} = \frac{4,000}{(\frac{5}{11})} = 44,000 \text{ Rs}$

iii) $MOS = \text{Sales} - BEP = 1,10,000 - 44,000$
 $= \text{Rs } 66,000$

② Decrease in fixed cost by 5000
 New F.C = 15,000

i) $\frac{P}{V} = \frac{S-V}{S} = \frac{1,00,000 - 60,000}{1,00,000} = \frac{2}{5}$

ii) $BEP = \frac{\text{Fixed cost}}{(\frac{P}{V})} = \frac{15,000}{\frac{2}{5}} = \frac{37,500}{2} = 37,500 \text{ Rs}$

iii) $MOS = \text{Sales} - BEP = 1,00,000 - 37,500$
 $= 62,500 \text{ Rs}$

④ The sales and profit of a company during two years was as follows:

years	sales (in Rs)	profit (in Rs)
2022	1,50,000	20,000
2023	1,70,000	25,000

you are required to calculate $\frac{P}{V}$ ratio, BEP, sales required to earn a profit of Rs 40,000, the profit made when sales are Rs 2,50,000, margin of safety, Variable cost.

Ans:- i) $\frac{P}{V} = \frac{\text{change in profit}}{\text{change in sales}} = \frac{5,000}{20,000} = \frac{1}{4} < 1$

ii) ~~BEP~~ considering the year 2023
 we know that $\frac{P}{V} = \frac{\text{contribution}}{\text{Sales}}$

$\frac{1}{4} = \frac{\text{Contribution}}{1,70,000}$

$\text{Contribution} = \frac{1,70,000}{4} = 42,500 \text{ Rs}$

But $\text{Contribution} = F + P$

$42,500 = F + 25,000$

$F = 17,500 \text{ Rs}$

$BEP = \frac{F}{(\frac{P}{V})} = \frac{17,500}{(\frac{1}{4})} = 70,000 \text{ Rs}$

iii) sales required to earn a profit of Rs 40,000

$\frac{FC + DP}{(\frac{P}{V})} = \frac{17,500 + 40,000}{(\frac{1}{4})} = 230,000 \text{ Rs}$

(iv) $2,50,000 = \frac{FC + DP}{(\frac{P}{V})}$

$2,50,000 = \frac{17,500 + DP}{(\frac{1}{4})}$

$\frac{2,50,000}{4} - 17,500 = DP = 45,000 \text{ Rs}$

(v) $MOS = \text{Sales} - BEP$

(2022) $= 1,50,000 - 70,000 = 80,000 \text{ Rs}$

$MOS (2023) = 1,70,000 - 70,000 = 1,00,000 \text{ Rs}$

vi) variable cost = $80 - \text{Contribution}$
 (2022) = ~~140,000~~
 $= S - F - P$
 $= 1,50,000 - 17,500 - 20,000$
 $= 1,12,500 \text{ Rs}$

(2023) $\Rightarrow 1,70,000 - 17,500 - 20,000$
 $= 1,27,500 \text{ Rs}$

⑩ the following figures relate a manufacturing company

	Total Sales	Total cost	profit
period 1	2,22,300	1,198,360	23,940
period 2	2,45,100	2,114,320	30,780

- Calculate
- Break even point
 - Fixed cost percentage to sales
 - Margin of safety
 - Variable cost

period-1

a) $BEP = \frac{\text{fixed cost}}{(\frac{P}{V}) \text{ ratio}}$

$\frac{P}{V} = \frac{198,360}{2,22,300} = \frac{1}{1.12}$

$\frac{P}{V} = \frac{\text{change in profits}}{\text{change in sales}} = \frac{23,940}{2,2800} = \frac{6840}{22800} = 0.3$

$\frac{P}{V} = \frac{\text{contribution}}{\text{sales}}$

$\frac{3}{10} = \frac{\text{contribution}}{222,300}$

contribution = 66,690 = F + P

66,690 - 23,940 = F

$F = 42,750$

$BEP = \frac{42,750}{(\frac{3}{10})} = 1,42,500 \text{ Rs}$

(b) $\frac{P}{\text{Sales}} \times 100 = \frac{42,750}{222,300} \times 100 = 19.23\%$

P-2
 $\frac{42,750}{245,100} \times 100 = 17.44\%$

(c) $MOS = \text{Total sales} - B.E.P$

$P_1(MOS) = 222,300 - 1,42,500$
 $= 79,800$

$P_2(MOS) = 245,100 - 1,42,500$
 $= 102,600$

d) $\text{Variable cost} = S - \text{Contribution}$
 (CP1)
 $= 222,300 - 66,690$
 $= 1,55,610 \text{ Rs}$

Variable cost = S - contribution
 (CP2)
 $= 2,45,100 - 73,530$
 $= 1,71,570 \text{ Rs}$

⑪ Suppose the break even sales are Rs 10,00,000 fixed costs are Rs 400,000 compute

(a) Contribution to sales ratio

(b) Selling price per unit if variable costs are Rs 12 per unit

(c) Margin of safety if 80,000 units are sold.

Sol: (a) $\frac{P}{V} = \frac{\text{fixed cost}}{BEP}$

$= \frac{10,00,000}{5,00,000} = 0.4$

(b) $\frac{P}{V} = \frac{\text{contribution}}{\text{sales}} = \frac{S_{pu} - V_{pu}}{\text{sales}}$

$0.4 = \frac{S_{pu} - 12}{S_{pu}} \Rightarrow$

$0.4 S_{pu} = S_{pu} - 12$

$0.6 S_{pu} = 12 \Rightarrow S_{pu} = 20 \text{ Rs}$

$$(c) \text{ mos} = \text{sales} - \text{BEP/unit}$$

$$= \frac{50,000}{1,00,000} = \frac{50,000}{100,000} = 50,000$$

$$= 80,000 - 50,000 = 30,000 \text{ units}$$

$$\text{mos (Rs)} = \text{Total sales} - \text{BEP}$$

$$= 80,000 \times 20 - 10,00,000$$

$$= 16,00,000 - 10,00,000$$

$$= 6,00,000 \text{ Rs}$$

(13) The following information is related to XYZ cement limited. Rs sales (10,000 tons Rs 5,00,000)

material cost - Rs 1,00,000

labour cost - Rs 50,000

direct expenses - Rs 10,000

fixed cost - Rs 3,00,000

profit - Rs 40,000

1,60,000

You are to find

(a) P ratio

(b) mos

(c) sales required to earn a profit of Rs 50,000

(d) profit at a sales level of Rs 7,00,000

$$\text{Ans: } \frac{P}{V} = \frac{\text{contribution}}{\text{sales}} = \frac{F + P}{\text{sales}} = \frac{3,00,000 + 40,000}{5,00,000}$$

$$= \frac{3,40,000}{5,00,000} = \frac{34}{50} = 0.68$$

$$\text{BEP} = \frac{F}{\left(\frac{P}{V}\right)} = \frac{40,000}{\frac{34}{50}} = \frac{10,00,000}{34}$$

$$= \frac{3,00,000 \times 25}{17} = 58823.52 \text{ Rs}$$

$$= \frac{75,00,000}{17} = 4,41,176.47$$

$$\begin{aligned} \text{mos} &= \text{Total sales} - \text{BEP} \\ &= 5,00,000 - 4,41,176.47 \\ &= 58823.53 \end{aligned}$$

$$\text{(A)} \quad \frac{F + P}{\left(\frac{P}{V}\right)} = \frac{3,00,000 + 50,000}{\left(\frac{17}{25}\right)} = \frac{3,50,000 \times 25}{17} = 5,14,705.88 \text{ Rs}$$

$$\begin{aligned} \text{(B)} \quad P &= S - V - F \\ &= 9,00,000 - 1,60,000 - 3,00,000 \\ &= 9,00,000 - 4,60,000 = 4,40,000 \end{aligned}$$

$$9,00,000 = \frac{FC + DP}{\frac{P}{V}}$$

$$DP = 3,12,000$$

Unit - 3 — Sold trader

Partnership Deal

The written agreement among the partners is called partnership deal. It contains the terms & conditions governing the working of partnership.

Contents of partnership deal

- ① Names & addresses of the firm and partners.
- ② Nature of business proposed
- ③ Duration
- ④ Amount of capital on the partnership and the ratio for contribution by each of the partners.
- ⑤ Their profit sharing ratio
- ⑥ Rate of interest charged on capital contributed, load taken from the partnership & the amounts drawn, if any, by the partners for their respective capital balances.
- ⑦ The amount of salary or commission payable to any partner.
- ⑧ Procedure to value good will of the firm at the time of admission of a new partner, retirement or death of a partner.
- ⑨ Allocation of responsibilities of the partners in the firm. Procedure for dissolution of the firm.
- ⑩ Name of the arbitrator to whom the disputes, if any can be referred to for settlement.
- ⑪ Special rights obligations, liabilities of partners if any

Joint stock company

An association of many persons who contribute money or moneys worth to a common stock and employed it for a common purpose

Features

- ① Artificial person separate
- ② Separate legal existence
- ③ Voluntary association of persons
- ④ Limited liability.
- ⑤ Capital is dividing into shares
- ⑥ Transferability of shares
- ⑦ Common seal
- ⑧ Perpetual succession
- ⑨ Ownership & management separated
- ⑩ winding up
- ⑪ The name of the company ends with 'limited'.

Advantages

- ① mobilisation of large resources
- ② Separate legal entity
- ③ limited liability
- ④ Transferability of shares
- ⑤ liquidity of investments
- ⑥ Encourage habit of savings & investments
- ⑦ Democracy in management.
- ⑧ Economy of larger scale production.
- ⑨ Continued existence
- ⑩ Institutional confidence
- Professional management
- Growth & expansion.

Dis

- ① Formation of company is a long drawn procedure.
- ② High degree of govt interference

- ② delayed decision making
- ③ lack of initiative
- ④ lack of responsibility & commitment